

OpenShift Single Node (SNO) Deployment – Production Runbook

1. Overview

This document describes the end-to-end procedure used to deploy a Single Node OpenShift (SNO) cluster in a disconnected environment using custom DNS, HAProxy, and a local mirror registry.

2. Environment Details

- Base Domain: `jaya.co`
 - Cluster Name: `ocp`
 - API Server: `api.ocp.jaya.co`
 - Ingress: `*.apps.ocp.jaya.co`
 - Network Plugin: `OVNKubernetes`
-

3. Install Configuration (install-config.yaml)

```
apiVersion: v1
baseDomain: jaya.co
metadata:
  name: ocp

compute:
- hyperthreading: Enabled
  name: worker
  replicas: 0

controlPlane:
  hyperthreading: Enabled
  name: master
  replicas: 1
```

```
networking:
  networkType: OVNKubernetes
  clusterNetwork:
    - cidr: 10.20.0.0/14
      hostPrefix: 23
  serviceNetwork:
    - 172.30.0.0/16

platform:
  none: {}

fips: false
pullSecret: ''
sshKey: ''
```

4. Pre-Requisites Setup

4.1 DNS Configuration (BIND)

Installed and configured BIND to resolve OpenShift cluster records:

```
/etc/named.conf
options {
    listen-on port 53 { 127.0.0.1; 192.168.96.119; };
    listen-on-v6 port 53 { ::1; };

    directory "/var/named";

    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file "/var/named/data/named.secroots";
    recursing-file "/var/named/data/named.recursing";

    allow-query { localhost; 192.168.0.0/16; any; };
};

zone "ocp.jaya.co" IN {
    type master;
    file "forward.zone";
    allow-update { none; };
};
```

```
};
```

```
zone "96.168.192.in-addr.arpa" IN {  
    type master;  
    file "reverse.zone";  
    allow-update { none; };  
};
```

```
/var/named/forward.zone
```

```
$TTL 1D
```

```
@ IN SOA bastion.ocp.jaya.co. root.ocp.jaya.co. (  
    2026041201  
    1D  
    1H  
    1W  
    3H )
```

```
@ IN NS bastion.ocp.jaya.co.
```

```
bastion    IN A 10.10.0.10  
api        IN A 10.10.0.10  
api-int    IN A 10.10.0.10  
*.apps     IN A 10.10.0.10  
master1    IN A 10.10.0.11
```

```
/var/named/reverse.zone
```

```
$TTL 1D
```

```
@ IN SOA bastion.ocp.jaya.co. root.ocp.jaya.co. (  
    2026041201  
    1D  
    1H  
    1W  
    3H )
```

```
@ IN NS bastion.ocp.jaya.co.
```

```
10 IN PTR bastion.ocp.jaya.co.  
11 IN PTR master1.ocp.jaya.co.
```

Required records:

- `api.ocp.jaya.co`
- `api-int.ocp.jaya.co`
- `*.apps.ocp.jaya.co`

Ensure proper forward and reverse resolution.

4.2 Load Balancer Configuration (HAProxy)

Configured HAProxy to route API and Ingress traffic:

```
/etc/haproxy/haproxy.cfg
global
    log          127.0.0.1 local2
    pidfile      /var/run/haproxy.pid
    maxconn      4000
    daemon

defaults
    mode tcp
    log global
    option dontlognull
    retries 3
    timeout connect 10s
    timeout client 1m
    timeout server 1m

frontend stats
    bind *:1936
    mode http
    stats enable
    stats uri /stats
    stats auth admin:jayaraj

# API Server
listen api-server-6443
    bind *:6443
    mode tcp
    server ocp-node 192.168.X.X:6443 check
```

```
# Machine Config Server (bootstrap phase only)
listen machine-config-server-22623
  bind *:22623
  mode tcp
  server ocp-node 192.168.X.X:22623 check

# Ingress 443
listen ingress-443
  bind *:443
  mode tcp
  server ocp-node 192.168.X.X:443 check

# Ingress 80
listen ingress-80
  bind *:80
  mode tcp
  server ocp-node 192.168.X.X:80 check
```

- Port 6443 → Kubernetes API
- Port 22623 → Machine Config Server
- Port 80/443 → Ingress router

4.3 Mirror Registry Setup

```
kind: ImageSetConfiguration
apiVersion: mirror.openshift.io/v1alpha2
archiveSize: 4
storageConfig:
  registry:
    imageURL: bastion.ocp.task.lab:8443/mirror/oc-mirror-metadata
    skipTLS: false
mirror:
  platform:
    channels:
      - name: stable-4.18
        type: ocp
        minVersion: 4.18.23
        maxVersion: 4.18.23
    graph: true
```

5. Installation Workflow

Step 1: Create Install Manifests

Generate manifests:

```
openshift-install create manifests
```

Step 2: Create Ignition Configs

```
openshift-install create ignition-configs
```

Generated files:

- `bootstrap.ign`
- `master.ign`

Step 3: Host Ignition File

Installed HTTPD server and placed ignition files:

```
cp bootstrap.ign /var/www/html/
```

Step 4: Boot Node (SNO Master)

- Booted the master node using RHCOS ISO

```
coreosinstaller install /dev/sda  
-ignition-url=http://192.168.96.119:8080/master.ign -insecure-ignition  
-copy-network -insecure
```

6. Post Installation Validation

Check cluster health:

```
oc get nodes  
oc get co  
oc get clusterversion
```

```
root@bastion:~/mirror
[root@bastion mirror]# oc get no
NAME                STATUS    ROLES    AGE    VERSION
master1.ocp.task.co Ready    control-plane,master,worker 134m    v1.31.11
[root@bastion mirror]# oc get co
NAME                VERSION    AVAILABLE    PROGRESSING    DEGRADED    SINCE    MESSAGE
authentication      4.18.23    True         False          False       99m
baremetal            4.18.23    True         False          False       126m
cloud-controller-manager 4.18.23    True         False          False       128m
cloud-credential     4.18.23    True         False          False       147m
cluster-autoscaler   4.18.23    True         False          False       126m
config-operator      4.18.23    True         False          False       126m
console              4.18.23    True         False          False       99m
control-plane-machine-set 4.18.23    True         False          False       126m
csi-snapshot-controller 4.18.23    True         False          False       126m
dns                  4.18.23    True         False          False       125m
etcd                 4.18.23    True         False          False       123m
image-registry        4.18.23    True         False          False       115m
ingress              4.18.23    True         False          False       126m
insights              4.18.23    True         False          False       103m
kube-apiserver        4.18.23    True         False          False       121m
kube-controller-manager 4.18.23    True         False          False       124m
kube-scheduler        4.18.23    True         False          False       123m
kube-storage-version-migrator 4.18.23    True         False          False       126m
machine-api           4.18.23    True         False          False       126m
machine-approver      4.18.23    True         False          False       126m
machine-config        4.18.23    True         False          False       124m
marketplace           4.18.23    True         False          False       125m
monitoring            4.18.23    True         False          False       99m
network               4.18.23    True         False          False       126m
node-tuning           4.18.23    True         False          False       126m
olm                   4.18.23    True         False          False       113m
openshift-apiserver   4.18.23    True         False          False       106m
openshift-controller-manager 4.18.23    True         False          False       125m
openshift-samples     4.18.23    True         False          False       115m
operator-lifecycle-manager 4.18.23    True         False          False       125m
operator-lifecycle-manager-catalog 4.18.23    True         False          False       125m
operator-lifecycle-manager-packageserver 4.18.23    True         False          False       121m
service-ca            4.18.23    True         False          False       126m
storage               4.18.23    True         False          False       126m
[root@bastion mirror]# oc get clusterversion
NAME    VERSION    AVAILABLE    PROGRESSING    SINCE    STATUS
version 4.18.23    True         False          99m      Cluster version is 4.18.23
[root@bastion mirror]#
```

Expected result:

- Single control-plane node in Ready state
- All Cluster Operators Available=True